

Exercise sheet 2

PRISM - Curves and Surfaces

For some of these exercises you will need to recall the arc length formula for a parametrization $\theta(t)$,

$$L(t) = \int_{t_0}^t \|\theta'(u)\| du$$

1. Consider a circle of radius r . Choose a parametrization of speed ν for it. Compute the magnitude of the acceleration of this parametrization.
2. Using the arc length formula introduced in the lecture, compute the arc length of a circle of radius r , any given line, and an ellipse. In each case you will also have to find a suitable parametrization.
3. Let $\theta_1(t)$ denote a parametrization defining a curve. If I define a new parametrization $\theta_2(t) = \theta_1(\tau(t))$, where τ is some reparametrization, then this is also another parametrization on the same curve. Choose a point p on the curve.
 - (a) How does the velocity of the first parametrization at point p relate with the velocity of the second parametrization at the same point p ?
 - (b) Show that if $\tau(t)$ is the arc length function introduced in the lecture, then the velocity of θ_2 would always have magnitude 1. Use the integral definition. Do not just describe it intuitively. Use calculus.